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To the Editors, *Journal of Chemical Information and Modeling*

Dear Editors,

We submit for your consideration an Article entitled **"Activity cliffs on human acetylcholinesterase: a pre-specified descriptor escalation"**.

It is widely held that two-dimensional fingerprints fail on activity cliffs because they cannot see the detail separating the two members of a pair, and that supplying it should help. We tested that directly by escalating four structure-based representations against the same fingerprint baseline on the same frozen splits, and we measured in advance what difference the design could resolve. On the development split none was resolvably better on any of three framings of the endpoint, and on per-ligand potency the molecular dynamics descriptors were resolvably worse. The result is bounded rather than open, which is what separates it from an underpowered null.

We also opened the sealed test split, once, after every descriptor set and algorithm had been fixed, and it does not reproduce that result. On the signed pair difference the receptor ensemble and the dynamics descriptors both lead the fingerprints there by about 0.3 in the coefficient of determination, and both gaps survive a correction for the twelve comparisons made and a leave-one-pair-out check. We report the disagreement rather than either half of it, because 38 held-out pairs against 145 development pairs cannot settle which reading is right. We would rather the Editors meet it here than find it in Section 3.5.

Three findings carry beyond this target. First, a replicate test separates persistence from amplitude sharply, so a trajectory of affordable length reproduces how long a contact is held and not how far things move. Second, a descriptor filter refitted at each value of a varied parameter manufactures a trend that a single fixed selection removes. This exposes any study that re-selects features as it varies a parameter. Third, the descriptors that survive reproducibility are available from 15 ns of simulation rather than the 50 the original design called for.

We would like to address three points of the Journal's guidance on QSAR submissions before a reader reaches them. First, this is not a docking study against a single target. The docking serves a comparison between representations, and the quantity under test is what physical detail adds to a matched ligand-based baseline. Second, the high ratio of descriptors to data points sits in the fingerprint baseline, scored as a practitioner would score it. We report a matched-dimensionality arm alongside it so that feature count can be separated from representation. Third, we are not proposing new descriptors and claim no superiority for them. What is offered is the comparison and the detection limit that makes its outcome readable, and the descriptor machinery is standard and attributed as such.

We chose acetylcholinesterase because the question needs a target supplying activity cliffs in quantity and a crystallographic human receptor, and because its gorge gates conformationally on evidence predating this work, so it is favourable to the hypothesis under test.

All molecular structures appear in the Supporting Information. All analysis code and derived data are in a public repository archived with a DOI, which is sufficient to reproduce every number without rerunning a simulation.

The manuscript is original, is not under consideration elsewhere, and has been approved by all authors. The authors declare no competing financial interest, and the work received no external funding.

Yours sincerely,

Isman Kurniawan, on behalf of all authors